EST NOTES

- Friction on Horizontal/Inclined Plane
  - Pulling load on surface
    - When applied force is pulled parallel to plane (even if plane is inclined/horizontal), no simultaneous equations needed.
    - When the applied force is not parallel, make Rn in terms of P, then solve for P by using $\Sigma F_x = 0$
  - Stacked Blocks (In Equilibrium)
    - When the system is at equilibrium, impending motion is in direction of force (unless specified otherwise)
    - All frictional forces (due to blocks stacked on top of it or block and surface) will oppose the impending motion
  - Determine Impending Motion (& Slipping)
    - First determine if the block is sliding by comparing θ and ϕ.
    - If the block is sliding, impeding motion will be down the slope (e.g. force needed to hold the block on the slope)
    - If the block is not sliding, impending motion will be specified.
- Friction on Screw Thread
  - Determining Formula Used (Against/Toward Axial Load)
    - Unless specified, the axial load is the force that would be exerted ON the screw.
    - e.g clamps $\longrightarrow$ axial load is normal contact of object being clamped on screw
    - turn buckle $\longrightarrow$ axial load is tension in rope pulling on screw
    - for turn buckle, after finding tension needed (T), the moment needed is Ttan(angles)(d/2)∗2 as 2 screws present
    - After determining axial load, determine the motion of the screw (should it be screwed in or out?)
    - if against, use formula $M = W\tan(\theta + \phi)\frac{d}{2}$. Else, use $M = W\tan(|\theta - \phi|)\frac{d}{2}$

- Linear Kinematics
  - Using SUVAT Equations
    - check what values are given to you
    - if time is not given at all (and not solving for it), use $v^2 = u^2 + 2as$
    - Otherwise, usually $s = ut + \frac{1}{2}at^2 = \frac{(v+u)}{2}t$ is used (make sure to change acceleration sign if needed for formula to work)
    - Initial displacement can be accounted by replacing s with $s - s_0$.
    - worse case use integration idk lol
  - Chasing
    - check that values are in m/s and not km/h ($x\ km/h = \frac{x}{3.6}\ m/s$)
    - CHECK that if chasing, do both start moving at same time or does one start moving before the other !!!!!!!!!!!!!!
    - graph may change in later parts of the question (e.g. max speed of A is now x m/s, when will it catch up with B)
    - Note that some questions may ask for ELAPSED not ABSOLUTE time. (i.e. time taken for a bike to reach X is not the same as time where a bike reaches X is the bike does not start at t=0.)
  - Free-fall
    - assuming no air resistance, when object thrown up has a = -9.81m/s^2 (slows down to 0 at max height) and will continue at same acceleration after max height, object thrown down has a = 9.81m/s^2 (will continue to speed up until hits ground)
    - account for initial velocity and displacement of object if there is !!!!
    - Graph will be like abs graph
  - make sure acceleration is presented correctly, -a acceleration or +a deceleration
- Angular Kinematics
  - Take note of rad/s and rev/min, where $x\ rev/min = \frac{2\pi}{60}x\ rad/s$
  - The general formula of

$$\frac{d^n s}{dt^n} = r\frac{d^n \theta}{dt^n}$$

is used for converting between linear and rotational displacement

Common Belts

for belts, every point on the belt has same linear velocity, thus $s_a = s_b$ (and other derivatives of time apply as well.)

this gives $r_a\theta_a = r_b\theta_b$ which can be formed into a ratio if needed (or just solved)

Common Axle

for common axles, both objects will rotate the same amount, thus $\theta_a = \theta_b$ (again, other derivatives of time apply)

Wounded String

General linear-angular conversion formula can be used to find descent of string, etc.

however if a weight is connected to string, wheel does not necessarily rotate to follow acceleration of gravity (or maybe it does under some conditions???)

Newtons Laws (non connected and connected)

DIRECTION OF NET FORCE & Acceleration

direction of net force will be same as acceleration

thus sign of sum should be the same (and net force must be added in same convention of acceleration)

Convention in freebody diagram should be in same direction as acceleration

USUALLY motion is specified $\longrightarrow$ acceleration direction is given $\longrightarrow$ net force must be summed in same direction

MAKE SURE TO READ QN ON SPECIFIED DIRECTION!!!!!!!!!

Incline Convention

Slopes may be written as incline of 1 in X, e.g. incline of 1 in 100, which refers to

$$\sin(\theta) = \frac{1}{100}$$

where theta is the angle of incline

Tractive effort and resistance

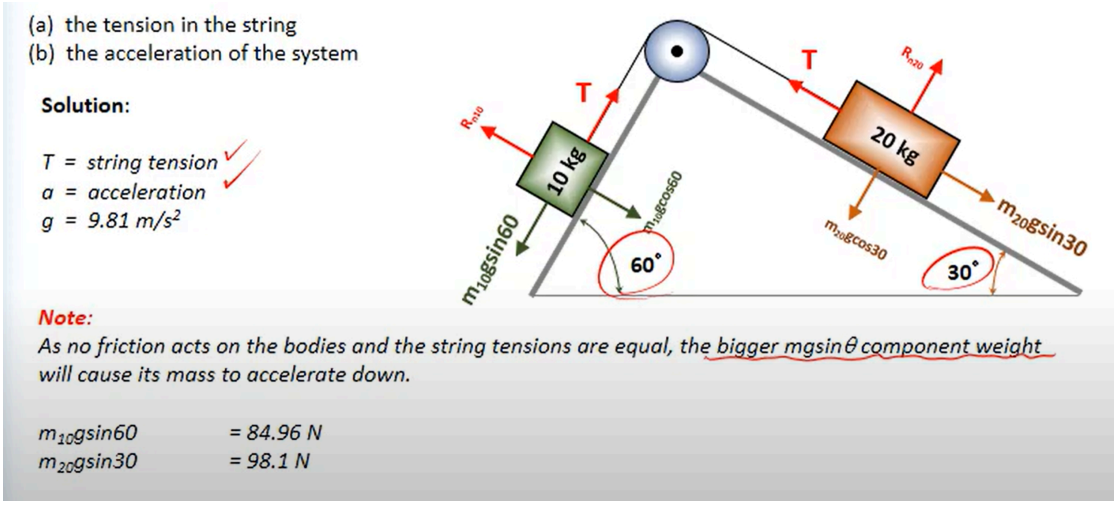
Tractive effort, F_TE is the engine effort used to drive the car (forward)
Tractive resistance, F_TR, is the net resistance from friction, air resistance, etc. This is often expressed as a rate of x/tonnes, which has to be calculated to find the actual tractive resistance.

Connected blocks on slopes

Usually, acceleration is in the direction of the heavier and steeper block. However, this may not always be apparent.

To systematically find the acceleration definition:

compare the mgsin(theta) components of both blocks (both are usually in different directions) and the greater one indicates acceleration direction



**note that even if you assume aceleration wrongly, acceleration will be negative to assumed direction

Connected blocks on slope + freefall

almost always, the block in freefall will be pulling in the direction of the acceleration, however this can be further verified by comparing mgsin(theta) and mg components (pulling components due to weight)

Double Pulley System

The main difference is that the tension of the block connected to the pulley is 2T, while the other block has T tension.

Acceleration is opposite, 2T block has acceleration a, T block has acceleration (minus) 2a,

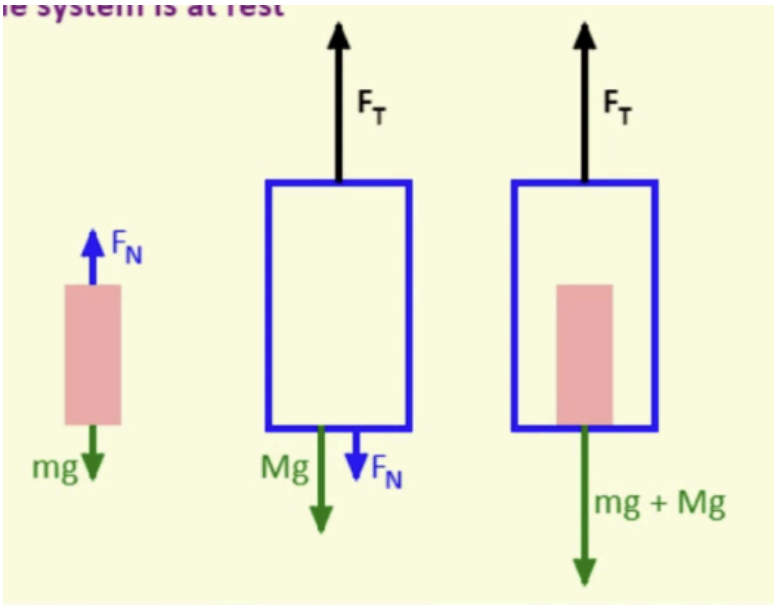
Elevator/truck bed/Caged Bodies

When one body is in another, the two are “connected” thus their accelerations will be the same.

For caged bodies moving vertically, the caged body has its weight acting on him, and the normal contact force of the cage on him acting on him as well (lets say that force is Rn)

The cage will have the applied force pulling it in the direction of acceleration, its own weight, and Rn, acting in the opposite direction.

note that since the system is accelerating, W_caged =/= Rn.



Thinking about this in terms of inertia, the caged will tend to stay still, but as the whole system is accelerating due to an applied force, Rn will force the caged to accelerate in the same direction. (wtf is this)

For caged bodies moving horizontally, friction is involved.

For the caged body, it will have only one horizontal force of friction in the direction of acceleration. This may seem like friction is in the same direction of motion, but if we consider the relative view of the moving cage, inertia would want to not move the caged body globally, thus from

the relative view, impending motion is opposite the direction of motion of the cage. This impending motion is what the friction opposes.

Main idea: consider all possible forces of each body, and see which ones would cause a net force to move the body in the direction of acceleration

Removing Tractive Force/Applied Force Deceleration & Displacement

When a tractive force/applied force/applied tension is removed, deceleration can be calculated similarly from F=ma, by considering all other forces except the applied force.

If the maximum velocity before the applied force removed can be found/is given, we can use the acceleration and the max velocity to find the displacement of the block (aka the inertia/whiplash kind of) with $v^2 = u^2 + 2as$.

Note that when something is decelerating, direction of acceleration will be opposite original moving direction